Fractagon Convergence Theorem

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1 Fractagon Convergence Theorem, Uniform Case

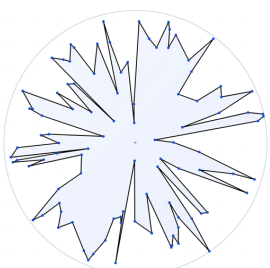
We consider the following geometric probability problem: sample n points uniformly at random from the closed unit disk $\mathbb{D} = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$, order them by polar angle, and connect consecutive points to form a polygon. We prove that the area of this *angularly induced polygon* (also called the **fractagon**) converges almost surely to $4\pi/9$ as $n \to \infty$:

Theorem 1.1 (Fractagon Convergence Theorem, Version 1)
Let A_n denote the signed area of the polygon formed by connecting n i.i.d. uniform points on the unit disk in angular order. Then

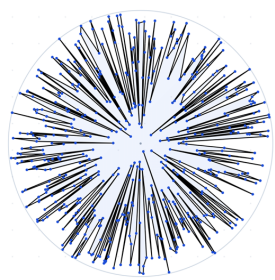
$$A_n \xrightarrow{\text{a.s.}} \frac{4\pi}{9} \approx 1.3963$$

as $n \to \infty$.

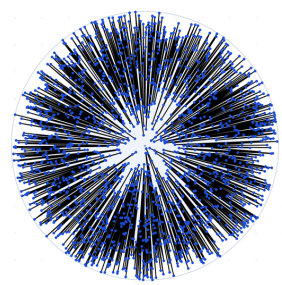
We will see that the convergence is driven by two phenomena: (1) the conditional expectation of the product of adjacent radii is always $\mathbb{E}[R]^2 = 4/9$, and (2) the sum $\sum_{i=1}^{n} \sin(\Delta\Theta_i)$ converges almost surely to 2π as the angular spacings become uniformly small.



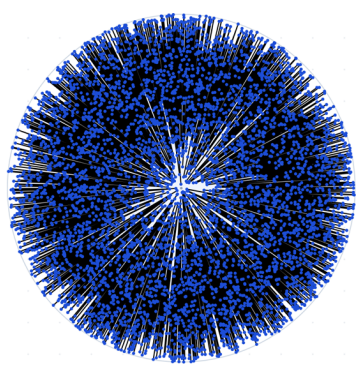
(a) Fractagon with 100 Vertices



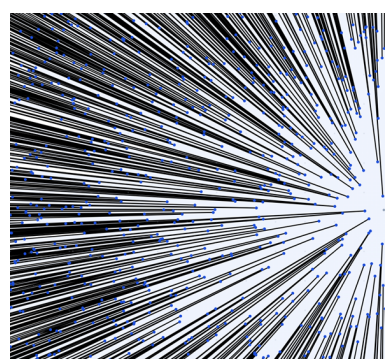
(b) Fractagon with 500 Vertices



(c) Fractagon with 2026 Vertices



(d) Fractagon with 5680 Vertices and Area 1.3987



(e) Detail of a Fractagon with 5680 Vertices

Figure 1: Samples of a fractagon with finitely many vertices.

1.1 Probability Space and Sampling Process

Definition 1.2 (Sampling Process) Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(Z_k)_{k\geq 1}$ be an i.i.d. sequence of random variables uniformly distributed on $\mathbb{D}$.

Each Z_k admits polar coordinates (R_k, Φ_k), where $R_k \in (0, 1]$ is the radial component and $\Phi_k \in [0, 2\pi)$ is the angular component. More formally:

Definition 1.3 (Polar Coordinate Map) Define the measurable map

$$T : \mathbb{R}^2 \setminus \{0\} \to (0, \infty) \times [0, 2\pi)$$

by

$$T(x, y) = \left(\sqrt{x^2 + y^2}, \arg(x + iy)\right),$$

where arg denotes the principal argument in $[0, 2\pi)$. For any random variable Z taking values in $\mathbb{D}$, we define polar coordinates $(R, \Phi) = T(Z)$ on the event $\{Z \neq 0\}$, with an arbitrary measurable extension to $\{Z = 0\}$.

Remark 1.4 The map T is Borel measurable. The polar coordinates (R, Φ) are therefore well-defined random variables regardless of whether Z admits a density with respect to Lebesgue measure. If $\mathbb{P}(Z = 0) = 0$, the arbitrary extension is irrelevant.

We recall some simple properties of the polar decomposition of the sequence $(Z_k)_{k\geq 1}$:

Proposition 1.5 (Marginal Distributions) The following properties hold:

(i) $\Phi_k \sim \text{Uniform}([0, 2\pi))$.

(ii) R_k has CDF $F_R(r) = r^2$ for $r \in [0, 1]$ and PDF $f_R(r) = 2r$.

(iii) R_k and Φ_k are independent.

Proof. For a uniform distribution on $\mathbb{D}$, the probability of landing in a measurable set $A \subseteq \mathbb{D}$ is $\mathbb{P}(Z \in A) = |A|/\pi$. The angular marginal is uniform by rotational symmetry. For the radial CDF,

$$F_R(r) = \mathbb{P}(R \leq r) = \frac{\pi r^2}{\pi} = r^2.$$

Independence follows from the product structure of the joint density in polar coordinates:

$$f_{R,\Phi}(r, \theta) = \frac{r}{\pi} = 2r \cdot \frac{1}{2\pi} = f_R(r) f_\Phi(\theta).$$

□

Corollary 1.6 (Moments of the Radius) Let R have PDF $f_R(r) = 2r$ on $[0, 1]$. Then

$$\mathbb{E}[R] = \frac{2}{3}, \qquad \mathbb{E}[R^2] = \frac{1}{2}, \qquad \text{Var}(R) = \frac{1}{18}.$$

Proof. This is a direct computation:

$$\mathbb{E}[R^k] = \int_0^1 r^k \cdot 2r\, dr = \frac{2}{k+2}.$$

□

1.2 Polygon Construction

Given a sample $Z_1, \ldots, Z_n$, we order the points by their angular coordinates to form a polygon:

Definition 1.7 (Order Statistics and Concomitants) Given a sample $(Z_1, \ldots, Z_n)$ with polar coordinates (R_k, Φ_k):

(i) Let

$$\Phi_{(1)} \leq \Phi_{(2)} \leq \cdots \leq \Phi_{(n)}$$

denote the order statistics of $(\Phi_1, \ldots, \Phi_n)$.

(ii) For each $i \in \{1, \ldots, n\}$, define the **concomitant** $R_{(i)}$ as the radius paired with $\Phi_{(i)}$: if $\Phi_{(i)} = \Phi_{\sigma(i)}$ for some permutation $\sigma \in S_n$, then $R_{(i)} = R_{\sigma(i)}$.

Definition 1.8 (Angularly Induced Polygon) The **n-th fractagon** $\mathcal{P}_n$ corresponding to a sample of $(Z_k)_{k\geq 1}$ is formed by connecting vertices in angular order:

$$(R_{(1)}, \Phi_{(1)}) \to (R_{(2)}, \Phi_{(2)}) \to \cdots \to (R_{(n)}, \Phi_{(n)}) \to (R_{(1)}, \Phi_{(1)}).$$

The following are the basic properties of the concomitants that we will use:

Lemma 1.9 (Properties of Concomitants) Conditionally on $\mathcal{G}_n := \sigma(\Phi_1, \ldots, \Phi_n)$, the random vector

$$(R_{(1)}, \ldots, R_{(n)})$$

is i.i.d. with the same law as $(R_1, \ldots, R_n)$. In particular:

(i) each $R_{(k)}$ has the same marginal distribution as R;

(ii) for distinct indices $i \neq j$, the random variables $R_{(i)}$ and $R_{(j)}$ are independent;

(iii) the vector $(R_{(1)}, \ldots, R_{(n)})$ is independent of the angular order statistics $(\Phi_{(1)}, \ldots, \Phi_{(n)})$.

Proof. Let σ be the random permutation induced by sorting the angles. Since the angles are independent of the radii, σ is independent of $(R_1, \ldots, R_n)$. Moreover,

$$(R_{(1)}, \ldots, R_{(n)}) = (R_{\sigma(1)}, \ldots, R_{\sigma(n)}).$$

Conditioned on $\mathcal{G}_n$, the permutation σ is deterministic, so $(R_{(1)}, \ldots, R_{(n)})$ is a deterministic permutation of an i.i.d. family and therefore has the same joint law as $(R_1, \ldots, R_n)$. The three claims follow immediately. □

Remark 1.10 (Non-Simple Polygons) For small n, the polygon $\mathcal{P}_n$ may self-intersect, as consecutive vertices in angular order need not be spatially adjacent. The area formula below computes the *signed (algebraic) area*, which remains well-defined and is the quantity studied in this note.

With the above definitions, we are now ready to obtain a formula for the area.

Definition 1.11 (Angular Spacings) Define the angular spacings:

$$\Delta\Theta_i = \begin{cases} \Phi_{(i+1)} - \Phi_{(i)}, & 1 \le i \le n-1, \\ 2\pi - \Phi_{(n)} + \Phi_{(1)}, & i = n. \end{cases}$$

Then $\Delta\Theta_i \ge 0$ almost surely and

$$\sum_{i=1}^{n} \Delta\Theta_i = 2\pi.$$

Proposition 1.12 (Shoelace Formula in Polar Coordinates) The signed area of $\mathcal{P}_n$ is

$$A_n = \frac{1}{2} \sum_{i=1}^{n} R_{(i)} R_{(i+1)} \sin(\Delta\Theta_i), \tag{1}$$

where $R_{(n+1)} := R_{(1)}$.

Proof. The polygon decomposes into n oriented triangles, each with one vertex at the origin and two consecutive vertices of $\mathcal{P}_n$. The signed area of the triangle with vertices at the origin, (r_1, θ_1), and (r_2, θ_2) is

$$\frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1).$$

Summing over all edges yields (1). □

1.3 Proof of the Main Result

Let us now prove the main result.

Theorem 1.13 (Fractagon Convergence Theorem, Version 1) As $n \to \infty$,

$$A_n \xrightarrow{\text{a.s.}} \frac{4\pi}{9}.$$

The proof proceeds in three steps: (1) compute the conditional expectation given the angles, (2) analyze the asymptotic behavior of the angular spacings, and (3) show that the centered fluctuation is summably small in fourth moment.

Lemma 1.14 (Step 1: Conditional Expectation)

$$\mathbb{E}[A_n \mid \mathcal{G}_n] = \frac{2}{9} \sum_{i=1}^{n} \sin(\Delta\Theta_i).$$

Proof. Conditioned on $\mathcal{G}_n$, the spacings $\Delta\Theta_i$ are deterministic. By Lemma 1.9, the concomitants $(R_{(i)})$ are i.i.d. with the same law as R and are independent of $\mathcal{G}_n$. Hence

$$\mathbb{E}[A_n \mid \mathcal{G}_n] = \frac{1}{2} \sum_{i=1}^{n} \sin(\Delta\Theta_i)\, \mathbb{E}[R_{(i)} R_{(i+1)} \mid \mathcal{G}_n].$$

For $n \geq 3$, adjacent concomitants $R_{(i)}$ and $R_{(i+1)}$ are distinct and independent, so by Corollary 1.6,

$$\mathbb{E}[R_{(i)} R_{(i+1)} \mid \mathcal{G}_n] = \mathbb{E}[R]^2 = \frac{4}{9}.$$

Substituting this into the previous display gives the result. □

We now analyze the sum

$$S_n := \sum_{i=1}^{n} \sin(\Delta\Theta_i).$$

Theorem 1.15 (Maximal Spacing) Let

$$M_n := \max_{1 \leq i \leq n} \Delta\Theta_i.$$

Then

$$\frac{n M_n}{\ln n} \xrightarrow{\text{a.s.}} 2\pi$$

as $n \to \infty$. In particular,

$$M_n \to 0 \qquad \text{almost surely.}$$

Proof. This is a classical result in the theory of uniform spacings; see [4, 3, 1]. □

Lemma 1.16 (Step 2: Sine Sum Convergence) The convergence

$$\sum_{i=1}^{n} \sin(\Delta\Theta_i) \xrightarrow{\text{a.s.}} 2\pi$$

holds as $n \to \infty$.

Proof. Since each $\Delta\Theta_i \geq 0$, we have

$$0 \leq \Delta\Theta_i - \sin(\Delta\Theta_i) \leq \frac{(\Delta\Theta_i)^3}{6}.$$

Therefore

$$0 \le 2\pi - \sum_{i=1}^{n} \sin(\Delta\Theta_i) = \sum_{i=1}^{n} \big(\Delta\Theta_i - \sin(\Delta\Theta_i)\big) \le \frac{1}{6} \sum_{i=1}^{n} (\Delta\Theta_i)^3.$$

Now

$$\sum_{i=1}^{n} (\Delta\Theta_i)^3 \le M_n^2 \sum_{i=1}^{n} \Delta\Theta_i = 2\pi M_n^2 \xrightarrow{\text{a.s.}} 0$$

by Theorem 1.15. Hence the claim follows. □

The previous lemma already shows that

$$\mathbb{E}[A_n \mid \mathcal{G}_n] \xrightarrow{\text{a.s.}} \frac{4\pi}{9}.$$

To conclude, we need to control the centered fluctuation.

Lemma 1.17 (Moments of the Squared Spacings) The normalized spacing vector

$$\left(\frac{\Delta\Theta_1}{2\pi}, \dots, \frac{\Delta\Theta_n}{2\pi} \right)$$

has the Dirichlet$(1, \dots, 1)$ distribution. Consequently,

$$\mathbb{E}\left[\sum_{i=1}^{n} (\Delta\Theta_i)^2 \right] = \frac{8\pi^2}{n+1},$$

and

$$\mathbb{E}\left[\left(\sum_{i=1}^{n} (\Delta\Theta_i)^2 \right)^2 \right] = \frac{64\pi^4 (n+5)}{(n+1)(n+2)(n+3)} = O\left(\frac{1}{n^2} \right).$$

Proof. The Dirichlet law of uniform spacings is classical; see, for instance, [5, 3]. If

$$X_i := \frac{\Delta\Theta_i}{2\pi},$$

then $(X_1, \dots, X_n) \sim$ Dirichlet$(1, \dots, 1)$, and the standard Dirichlet moment formulas give

$$\mathbb{E}[X_i^2] = \frac{2}{n(n+1)}, \qquad \mathbb{E}[X_i^4] = \frac{24}{n(n+1)(n+2)(n+3)},$$

and, for $i \neq j$,

$$\mathbb{E}[X_i^2 X_j^2] = \frac{4}{n(n+1)(n+2)(n+3)}.$$

Therefore

$$\mathbb{E}\left[\sum_{i=1}^{n} (\Delta\Theta_i)^2 \right] = (2\pi)^2 n \, \mathbb{E}[X_1^2] = \frac{8\pi^2}{n+1},$$

and

$$\begin{aligned}\mathbb{E}\left[\left(\sum_{i=1}^{n}(\Delta\Theta_i)^2\right)^2\right] &= (2\pi)^4\left(n\,\mathbb{E}[X_1^4]+n(n-1)\mathbb{E}[X_1^2X_2^2]\right)\\ &= (2\pi)^4\left(\frac{24n+4n(n-1)}{n(n+1)(n+2)(n+3)}\right)\\ &= \frac{64\pi^4(n+5)}{(n+1)(n+2)(n+3)}.\end{aligned}$$

□

Lemma 1.18 (Step 3: Fourth-Moment Concentration) There exists a constant $C>0$ such that

$$\mathbb{E}\Big[|A_n-\mathbb{E}[A_n\mid\mathcal{G}_n]|^4\,\Big|\,\mathcal{G}_n\Big]\le C\left(\sum_{i=1}^{n}\sin^2(\Delta\Theta_i)\right)^2\le C\left(\sum_{i=1}^{n}(\Delta\Theta_i)^2\right)^2$$

almost surely. In particular,

$$\mathbb{E}\Big[|A_n-\mathbb{E}[A_n\mid\mathcal{G}_n]|^4\Big]=O\left(\frac{1}{n^2}\right).$$

Proof. Set

$$Y_i:=R_{(i)}R_{(i+1)}-\frac{4}{9},\qquad b_i:=\frac{1}{2}\sin(\Delta\Theta_i),$$

so that

$$A_n-\mathbb{E}[A_n\mid\mathcal{G}_n]=\sum_{i=1}^{n}b_iY_i.$$

Conditionally on $\mathcal{G}_n$, the sequence $(Y_i)_{i=1}^n$ is centered, bounded, and 1-dependent, because Y_i depends only on $R_{(i)}$ and $R_{(i+1)}$. A standard fourth-moment inequality for centered m-dependent sequences then yields

$$\mathbb{E}\left[\left|\sum_{i=1}^{n}b_iY_i\right|^4\,\middle|\,\mathcal{G}_n\right]\le C\left(\sum_{i=1}^{n}b_i^2\right)^2$$

for a constant C depending only on the law of R; see [2, Chapter 18]. Since $b_i^2\le\frac{1}{4}\sin^2(\Delta\Theta_i)\le\frac{1}{4}(\Delta\Theta_i)^2$, the stated bound follows. Taking expectations and applying Lemma 1.17 gives the final estimate. □

Proof of Theorem 1.13. By Lemmas 1.14 and 1.16,

$$\mathbb{E}[A_n\mid\mathcal{G}_n]\xrightarrow{\text{a.s.}}\frac{4}{9}\cdot\frac{2\pi}{2}=\frac{4\pi}{9}.$$

On the other hand, Lemma 1.18 implies

$$\sum_{n=1}^{\infty}\mathbb{P}(|A_n-\mathbb{E}[A_n\mid\mathcal{G}_n]|>\varepsilon)\le\frac{1}{\varepsilon^4}\sum_{n=1}^{\infty}\mathbb{E}\Big[|A_n-\mathbb{E}[A_n\mid\mathcal{G}_n]|^4\Big]<\infty.$$

By the Borel–Cantelli lemma,

$$A_n - \mathbb{E}[A_n \mid \mathcal{G}_n] \xrightarrow{\text{a.s.}} 0.$$

Combining the last two displays yields

$$A_n \xrightarrow{\text{a.s.}} \frac{4\pi}{9}.$$

□

2 Fractagon Convergence Theorem, Version 2

We now generalize the fractagon convergence theorem to distributions on the unit disk where the radial and angular components may be dependent. The limiting area is governed by the L^2-norm of the conditional expectation of the radius given the angle.

Throughout this section, we keep the notation introduced above for order statistics, concomitants, and angular spacings.

- **Sampling Process:** Let $(Z_k)_{k\geq 1}$ be a sequence of independent and identically distributed random variables taking values in $\mathbb{D}$. Write (R_k, Φ_k) for the polar coordinates.
- **Uniform Angular Marginal:** The angular component Φ has the uniform distribution on $[0, 2\pi)$.
- **Regression Function:** The conditional expectation of the radius given the angle,

  $$m(\theta) := \mathbb{E}[R \mid \Phi = \theta],$$

  is represented by a bounded Lipschitz function on $[0, 2\pi)$.

Remark 2.1 Since $R \in [0, 1]$ almost surely, we automatically have $\|m\|_\infty \leq 1$. The Lipschitz assumption is stronger than mere almost-everywhere continuity, but it is sufficient for a clean and rigorous proof.

Definition 2.2 (Second Conditional Moment) Define the conditional second moment function

$$\mu_2(\theta) := \mathbb{E}[R^2 \mid \Phi = \theta].$$

Since $R \leq 1$, we have $\mu_2(\theta) \leq 1$ for almost every θ.

We seek to prove the following result:

Theorem 2.3 (Generalized Fractagon Area Convergence)
Under the above assumptions, the area A_n converges almost surely:

$$A_n \xrightarrow{\text{a.s.}} \frac{1}{2}\int_0^{2\pi} m(\theta)^2 \, d\theta = \frac{1}{2}\|m\|^2_{L^2([0,2\pi))}.$$

Remark 2.4 (Recovery of the Independent Case) If R and Φ are independent, then $m(\theta) = \mathbb{E}[R]$ is constant, and

$$\lim_{n\to\infty} A_n = \frac{1}{2} \cdot (\mathbb{E}[R])^2 \cdot 2\pi = \pi(\mathbb{E}[R])^2 \quad \text{a.s.}$$

For the uniform distribution on $\mathbb{D}$, where $\mathbb{E}[R] = 2/3$, this yields $A_n \to 4\pi/9$.

2.1 Conditional Independence Structure

The key point is to understand the conditional distribution of the radii given the angles.

Lemma 2.5 (Conditional Independence of Radii) Conditional on $\mathcal{G}_n := \sigma(\Phi_1, \ldots, \Phi_n)$, the random variables $R_1, \ldots, R_n$ are independent. More precisely, for any Borel sets $B_1, \ldots, B_n \subseteq [0, 1]$,

$$\mathbb{P}(R_1 \in B_1, \ldots, R_n \in B_n \mid \mathcal{G}_n) = \prod_{k=1}^{n} \mathbb{P}(R \in B_k \mid \Phi = \Phi_k) \quad \text{a.s.}$$

Proof. Because the pairs (R_k, Φ_k) are i.i.d., the conditional law of $(R_1, \ldots, R_n)$ given $(\Phi_1, \ldots, \Phi_n)$ factors coordinatewise. This is exactly the displayed formula. □

The following is the corresponding analogue of Lemma 1.9:

Lemma 2.6 (Properties of Concomitants) Conditional on $\mathcal{G}_n$:

(i) the concomitant $R_{(i)}$ has the conditional distribution of R given $\Phi = \Phi_{(i)}$;

(ii) for distinct indices $i \neq j$, the concomitants $R_{(i)}$ and $R_{(j)}$ are conditionally independent;

(iii) for every i,

$$\mathbb{E}[R_{(i)} \mid \mathcal{G}_n] = m(\Phi_{(i)}), \qquad \mathbb{E}[R_{(i)}^2 \mid \mathcal{G}_n] = \mu_2(\Phi_{(i)}).$$

Proof. The permutation that orders the angles is $\mathcal{G}_n$-measurable. Since $R_{(i)}$ is just the radius attached to $\Phi_{(i)}$, the claims follow immediately from Lemma 2.5. □

Proposition 2.7 (Conditional Product Moments) For distinct indices $i, j \in \{1, \ldots, n\}$,

$$\mathbb{E}[R_{(i)} R_{(j)} \mid \mathcal{G}_n] = m(\Phi_{(i)})\, m(\Phi_{(j)}) \quad \text{a.s.}$$

Proof. By conditional independence,

$$\mathbb{E}[R_{(i)} R_{(j)} \mid \mathcal{G}_n] = \mathbb{E}[R_{(i)} \mid \mathcal{G}_n]\, \mathbb{E}[R_{(j)} \mid \mathcal{G}_n].$$

Now apply Lemma 2.6. □

Lemma 2.8 (Conditional Expectation Shoelace Formula)

$$\mathbb{E}[A_n \mid \mathcal{G}_n] = \frac{1}{2}\sum_{i=1}^{n} m(\Phi_{(i)})\, m(\Phi_{(i+1)})\, \sin(\Delta\Theta_i),$$

where $\Phi_{(n+1)} := \Phi_{(1)}$.

Proof. Conditioning the shoelace formula (1) and applying Proposition 2.7 gives

$$\mathbb{E}[A_n \mid \mathcal{G}_n] = \frac{1}{2}\sum_{i=1}^{n} \mathbb{E}[R_{(i)}R_{(i+1)} \mid \mathcal{G}_n]\, \sin(\Delta\Theta_i) = \frac{1}{2}\sum_{i=1}^{n} m(\Phi_{(i)})\, m(\Phi_{(i+1)})\, \sin(\Delta\Theta_i).$$

□

2.2 Convergence of the Conditional Expectation

We now show that $\mathbb{E}[A_n \mid \mathcal{G}_n]$ converges to the target integral.

Proposition 2.9 (Convergence of Conditional Expectation) Let

$$\mu := \frac{1}{2}\int_0^{2\pi} m(\theta)^2\, d\theta.$$

Then

$$\mathbb{E}[A_n \mid \mathcal{G}_n] \xrightarrow{\text{a.s.}} \mu.$$

Moreover, there exists a constant $C > 0$ such that

$$|\mathbb{E}[A_n \mid \mathcal{G}_n] - \mu| \le C\sum_{i=1}^{n}(\Delta\Theta_i)^2 \quad \text{a.s.}$$

Proof. Write

$$m_i := m(\Phi_{(i)}).$$

By Lemma 2.8,

$$\mathbb{E}[A_n \mid \mathcal{G}_n] = \frac{1}{2}\sum_{i=1}^{n} m_i m_{i+1} \sin(\Delta\Theta_i).$$

We compare this to μ in three steps.

Step 1: replace $\sin(\Delta\Theta_i)$ by $\Delta\Theta_i$. Since $0 \le x - \sin x \le x^3/6$ for $x \ge 0$,

$$\left|\sum_{i=1}^{n} m_i m_{i+1} \sin(\Delta\Theta_i) - \sum_{i=1}^{n} m_i m_{i+1}\Delta\Theta_i\right| \le \|m\|_\infty^2 \sum_{i=1}^{n} \frac{(\Delta\Theta_i)^3}{6}.$$

Because

$$\sum_{i=1}^{n}(\Delta\Theta_i)^3 \le M_n \sum_{i=1}^{n}(\Delta\Theta_i)^2,$$

this contribution is bounded by a constant times $\sum_i(\Delta\Theta_i)^2$.

Step 2: replace $m_i m_{i+1}$ by m_i^2. If $L := \mathrm{Lip}(m)$, then

$$|m_{i+1} - m_i| \le L\,\Delta\Theta_i,$$

hence

$$\left|\sum_{i=1}^{n} m_i m_{i+1}\Delta\Theta_i - \sum_{i=1}^{n} m_i^2\Delta\Theta_i\right| \le \|m\|_\infty L \sum_{i=1}^{n}(\Delta\Theta_i)^2.$$

Step 3: compare the remaining Riemann sum with the integral. The function $f := m^2$ is Lipschitz, with

$$\mathrm{Lip}(f) \le 2\|m\|_\infty L.$$

For each interval $[\Phi_{(i)}, \Phi_{(i+1)}]$, we therefore have

$$\left|f(\Phi_{(i)})\Delta\Theta_i - \int_{\Phi_{(i)}}^{\Phi_{(i+1)}} f(\theta)\,d\theta\right| \le \mathrm{Lip}(f)\,(\Delta\Theta_i)^2.$$

Summing over i gives

$$\left|\sum_{i=1}^{n} m_i^2\Delta\Theta_i - \int_0^{2\pi} m(\theta)^2\,d\theta\right| \le \mathrm{Lip}(f)\sum_{i=1}^{n}(\Delta\Theta_i)^2.$$

Putting the three estimates together yields

$$|\mathbb{E}[A_n \mid \mathcal{G}_n] - \mu| \le C\sum_{i=1}^{n}(\Delta\Theta_i)^2$$

for a suitable constant $C > 0$. Finally,

$$\sum_{i=1}^{n}(\Delta\Theta_i)^2 \le M_n \sum_{i=1}^{n}\Delta\Theta_i = 2\pi M_n \xrightarrow{\text{a.s.}} 0$$

by Theorem 1.15, and the almost sure convergence follows. □

2.3 Almost Sure Convergence

As in Version 1, it remains to show that A_n concentrates around its conditional expectation.

Lemma 2.10 (Fourth-Moment Concentration) There exists a constant $C > 0$ such that

$$\mathbb{E}\Big[|A_n - \mathbb{E}[A_n \mid \mathcal{G}_n]|^4 \,\Big|\, \mathcal{G}_n\Big] \le C\left(\sum_{i=1}^{n} \sin^2(\Delta\Theta_i)\right)^2 \le C\left(\sum_{i=1}^{n} (\Delta\Theta_i)^2\right)^2$$

almost surely. In particular,

$$\mathbb{E}\Big[|A_n - \mathbb{E}[A_n \mid \mathcal{G}_n]|^4\Big] = O\left(\frac{1}{n^2}\right).$$

Proof. Set

$$c_i := \frac{1}{2}\sin(\Delta\Theta_i), \qquad W_i := R_{(i)}R_{(i+1)} - m(\Phi_{(i)})m(\Phi_{(i+1)}).$$

Then

$$A_n - \mathbb{E}[A_n \mid \mathcal{G}_n] = \sum_{i=1}^{n} c_i W_i.$$

Conditionally on $\mathcal{G}_n$, the sequence (W_i) is centered, bounded, and 1-dependent. Therefore the same fourth-moment inequality used in Lemma 1.18 applies:

$$\mathbb{E}\left[\left|\sum_{i=1}^{n} c_i W_i\right|^4 \,\middle|\, \mathcal{G}_n\right] \le C\left(\sum_{i=1}^{n} c_i^2\right)^2$$

for some constant $C > 0$; see [2, Chapter 18]. Since $c_i^2 \le \frac{1}{4}(\Delta\Theta_i)^2$, the displayed bound follows. Taking expectations and using Lemma 1.17 gives the unconditional estimate. □

We are now ready to prove the theorem.

Proof of Theorem 2.3. By Proposition 2.9,

$$\mathbb{E}[A_n \mid \mathcal{G}_n] \xrightarrow{\text{a.s.}} \mu, \qquad \mu = \frac{1}{2}\int_0^{2\pi} m(\theta)^2\, d\theta.$$

By Lemma 2.10,

$$\sum_{n=1}^{\infty} \mathbb{P}(|A_n - \mathbb{E}[A_n \mid \mathcal{G}_n]| > \varepsilon) < \infty.$$

Hence, by Borel–Cantelli,

$$A_n - \mathbb{E}[A_n \mid \mathcal{G}_n] \xrightarrow{\text{a.s.}} 0.$$

Combining the last two displays yields

$$A_n \xrightarrow{\text{a.s.}} \mu.$$

□

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